Python Engineering - Class 02 Theory Guide

Conditionals and Special Cases

This guide explains the programming, mathematical and/or engineering theory behind the exercise. The goal is to understand the model before memorizing syntax.

1. Decision trees

Conditionals transform a single linear sequence into branches. Engineering problems often contain regions, limits or invalid domains.

Relationship / equation

```
condition -> branch A / branch B / branch C
```

Connection to code

The code should order conditions so that mutually exclusive cases are clear and unreachable code is avoided.

2. Rocket equation domain

The logarithm requires a positive mass ratio and the physical model assumes positive masses. For a propulsive burn m0 must be larger than mf.

Relationship / equation

```
Delta_v = Isp*g0*ln(m0/mf)
```

Connection to code

If m0=mf then ln(1)=0; if mf<=0 the mathematical operation becomes invalid.

3. Quadratic discriminant

The discriminant determines the root structure of a quadratic equation.

Relationship / equation

```
D=b^2-4ac
```

Connection to code

D>0 gives two real roots, D=0 one repeated root, D<0 a complex-conjugate pair. a=0 is not quadratic and must be treated separately.

4. Piecewise models

Many engineering correlations are defined by regions. A piecewise function is naturally represented using ordered conditional branches.

Relationship / equation

```
Cd(M) = region-dependent
```

Connection to code

The elif M<1.2 branch implicitly means the previous M<0.8 condition was false, therefore 0.8<=M<1.2.

Manual validation method

For any exercise in this class, use the following protocol before trusting the output:

- 1. Write input values with units or types.
- 2. Predict sign, scale, type and shape of the result.
- 3. Manually calculate a small case when possible.
- 4. Run the code and compare against the prediction.
- 5. Change one input in a controlled way and explain the effect.

Connection to the rest of the course

Conditionals will later define numerical convergence rules, data validity, flight regimes and exception paths.

Final summary

```
MODEL / CONCEPT
      |
      v
ALGORITHM
      |
      v
PYTHON IMPLEMENTATION
      |
      v
REFERENCE CHECK
      |
      v
ENGINEERING INTERPRETATION
```

> Syntax is the tool. The transferable skill is being able to model, structure, validate and interpret.